Estimators, Estimation and Sampling

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Outline

- Definition of estimator
- Biased and unbiased estimators
- Sampling and estimators
- Maximum Likelihood Estimation
- Maximum A Posteriori Estimation

Definition of estimator

- Given a data set $\mathcal{D}$ of m samples in $\mathbf{x}^{(i)} \in \mathbb{R}^d$
- Samples $\mathbf{x}^{(i)}$ can be seen as random manifestation of the phenomenon they described, represented from the random variable $\mathbf{x}$
- The foundational hypothesis of machine learning over $\mathbf{x}^{(i)}$ samples is that they are statistically independent and identically distributed (i.i.d. hypothesis) from a given probability distribution.

Definition of estimator

- An estimator or a statistic is a function of our data which tries to provide the *best possible prediction* of a quantity or function we are interested in
- Point estimation of a parameter $\boldsymbol{\theta}$ as a function of the data $\mathbf{x}^{(i)}$

$$\hat{\boldsymbol{\theta}}_m = g(\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)})$$

- Estimates are themselves random variables

Definition of estimator

- Estimation of a variable $\boldsymbol{y}$ as a *function* of the data

$$\boldsymbol{y} = f(\boldsymbol{x}) + \boldsymbol{\varepsilon}$$

- The relationship between $\boldsymbol{y}$ and $\boldsymbol{x}$ is not completely described by f
- We estimate a function f that approximates the true function f^*
- It is a *point* estimate in the function space

Biased and unbiased estimators

- The bias of an estimator is measured as the difference between its expected value and the true value of the quantity to be estimated

$$\mathrm{bias}(\hat{\boldsymbol{\theta}}_m) = \mathbb{E}\left[\hat{\boldsymbol{\theta}}_m\right] - \boldsymbol{\theta}$$

- It depends on the data we have to calculate the estimate

Biased and unbiased estimators

- Unbiased Estimator
  - An estimator whose expected value equals the true parameter value

$$\mathbb{E}\left[\hat{\boldsymbol{\theta}}_m\right] = \boldsymbol{\theta}$$

- Asymptotically Unbiased Estimator
  - An estimator whose bias approaches zero as the sample size increases to infinity

$$\lim_{m \to \infty} \mathbb{E}\left[\hat{\boldsymbol{\theta}}_m\right] = \boldsymbol{\theta}$$

Sample estimators of mean and variance

- Let assume that samples are drawn from a normal distribution

$$\forall i,\ x^{(i)} \sim \mathcal{N}(x; \mu, \sigma)$$

Sample mean $$\hat{\mu}_m = \frac{1}{m}\sum_{i=1}^{m} x^{(i)}$$ unbiased

Sample estimators of mean and variance

- Let assume that samples are drawn from a normal distribution

$$\forall i,\ x^{(i)} \sim \mathcal{N}(x; \mu, \sigma)$$

Sample variance

$$\hat{\sigma}_m^2 = \frac{1}{m}\sum_{i=1}^{m}(x^{(i)} - \hat{\mu}_m)^2 \qquad \text{biased}$$

$$\tilde{\sigma}_m^2 = \frac{1}{m-1}\sum_{i=1}^{m}\left(x^{(i)} - \hat{\mu}_m\right)^2 \qquad \text{unbiased}$$

Variance of an estimator

- The variance of an estimator is the measure used to determine how stable the estimate is with respect to the variation of the used samples
- The square root of variance is defined as standard error

$$\mathrm{SE}(\hat{\theta}_m) = \sqrt{\mathrm{Var}(\hat{\theta}_m)}$$

Standard error of the sample mean

$$\mathrm{SE}\left(\hat{\mu}_m\right) = \sqrt{\mathrm{Var}\left[\frac{1}{m}\sum_{i=1}^{m} x^{(i)}\right]} = \frac{\sigma}{\sqrt{m}}$$

Mean Square Error (MSE)

- The ideal estimator has minimum bias, better if it is unbiased, and minimum variance.
- To judge the goodness of an estimate in this sense, the Mean Square Error (MSE) is used.

$$\mathrm{MSE} = \mathbb{E}\left[(\hat{\theta}_m - \theta)^2\right] =$$

$$\mathrm{Bias}(\hat{\theta}_m)^2 + \mathrm{Var}(\hat{\theta}_m)$$

Monte Carlo Sampling and Analysis

- The estimators seen so far are simple and fall into the family of those called Monte Carlo

- Monte Carlo methods are computational methods that use *random sampling* to return a numerical estimate
  - They can be useful for overcoming computationally complex problems

- In this context, sampling, i.e. the choice of samples from which derive the estimate, is essential

Monte Carlo Sampling and Analysis

- In the Monte Carlo simulation of our interest, data are sampled according to a certain distribution
- The idea is to is to view the sum as if it were an expectation under some distribution and to approximate the expectation by a corresponding average

$$s = \sum_{\boldsymbol{x}} p(\boldsymbol{x}) f(\boldsymbol{x}) = E_p[f(\mathbf{x})]$$

Probability distribution

Monte Carlo Sampling and Analysis

- In the Monte Carlo simulation of our interest, data are sampled according to a certain distribution to approximate a numerically intractable integral with a finite series of sums
- This integral can be seen as the *calculation of an expected value* estimated through the corresponding mean

$$s = \int p(\boldsymbol{x}) f(\boldsymbol{x}) d\boldsymbol{x} = \mathbb{E}_p \left[f(\mathbf{x}) \right]$$

Probability density

Monte Carlo Sampling and Analysis

- We can approximate *s* by drawing *n* samples $\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}$ from *p*, and then forming the empirical average

$$\hat{s}_n = \frac{1}{n} \sum_{i=1}^{n} f(\boldsymbol{x}^{(i)})$$

$$\mathbb{E}[\hat{s}_n] = \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}[f(\boldsymbol{x}^{(i)})] = \frac{1}{n} \sum_{i=1}^{n} s = s$$

which results to be *unbiased*

Monte Carlo Sampling and Analysis

Sampling with uniform distribution of samples

- It provides the estimators seen previously
- The variance estimate is biased
- In this case $p(x) = \frac{1}{N}$

$$s = \int p(\boldsymbol{x}) f(\boldsymbol{x}) d\boldsymbol{x}$$

Monte Carlo Sampling and Analysis

Stratified sampling

- The population is divided into L non-overlapping partitions called strata
- Each stratum has its own portion S_k of the population
- From each stratum, n_k are drawn

Monte Carlo Sampling and Analysis

Stratified sampling

It represents the percentage of the population that falls within the stratum

$$\hat{\mu} = \sum_{k=1}^{L} \frac{S_k}{S} \mu_k, \quad \hat{\sigma}^2 = \sum_{k=1}^{L} \sigma_k^2,$$

The data has been sampled independently in each stratum.
The sample variance is therefore the total variance of the data, i.e. is the sum of the individual stratum variances

Monte Carlo Sampling and Analysis

Importance Sampling

- The basic idea is to approximate the probability distribution for which we want to calculate the expected value
  - the integral that is the true objective of the estimation
- The distribution is not known

Monte Carlo Sampling and Analysis

Importance Sampling

- We introduce a new probability distribution, *known*, from which to draw samples
- We call this distribution importance function because it conditions the choice of samples that are drawn from it

$$s = \int p(\boldsymbol{x}) f(\boldsymbol{x}) d\boldsymbol{x} = \int q(\boldsymbol{x}) \frac{p(\boldsymbol{x}) f(\boldsymbol{x})}{q(\boldsymbol{x})} d\boldsymbol{x}$$

$$s = \mathbb{E}_{\mathbf{x} \sim q} \left[\frac{pf}{q} \right]$$

Monte Carlo Sampling and Analysis

Importance Sampling

$$\hat{s}_p = \frac{1}{n} \sum_{i=1, \mathbf{x}^{(i)} \sim p}^{n} f(\boldsymbol{x}^{(i)})$$

$$\hat{s}_q = \frac{1}{n} \sum_{i=1, \mathrm{x}^{(i)} \sim q}^{n} \frac{p\left(\boldsymbol{x}^{(i)}\right) f\left(\boldsymbol{x}^{(i)}\right)}{q\left(\boldsymbol{x}^{(i)}\right)}$$

Maximum Likelihood Estimation

The Maximum Likelihood Estimantion (MLE)

- is one of the principles that can be used to define whether a particular function *represents a good estimator for a certain class of learning models*
- is one of the founding principles of machine learning
- if, given the problem to solve, it is possible to identify the most suitable model, MLE provides a criterion to estimate the error made and the reliability of the prediction.

Maximum Likelihood Estimation

The Maximum Likelihood Estimantion (MLE)

- is a method of estimating the parameters of a probability distribution by maximizing a likelihood function
- the objective is to find the parameter values that make the observed data most probable under the assumed statistical model.

Maximum Likelihood Estimation

- MLE is one of the principles which can be used to determine if one particular function represents a good estimator for a given class of learning modesl
- MLE assumes that our model maximizes its probability $P(\mathcal{D}, \mathbf{w})$ only with the maximization of $P(\mathcal{D} \mid \mathbf{w})$

- The estimation will be an empirical esitamtion since we own only n samples $\mathbf{x}^{(i)}$ of $\mathbf{x}$, that is the data set $\mathcal{D}$.

Maximum Likelihood Estimation

$$\mathbb{X} = \{\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}\}, \; p_{\text{data}}(\mathbf{x})$$ *unknown*

$$\boldsymbol{\theta}_{\text{ML}} = \arg\max_{\boldsymbol{\theta}} \; p_{\text{model}} \left(\mathbb{X}; \boldsymbol{\theta}\right)$$

$$= \arg\max_{\boldsymbol{\theta}} \prod_{i=1}^{m} p_{\text{model}} \left(\boldsymbol{x}^{(i)}; \boldsymbol{\theta}\right)$$

Assumption of statistical independence of samples, allows to express the joint probability as a product of individual probabilities for each sample

Maximum Likelihood Estimation

take logarithms: we obtain a summation

$$\boldsymbol{\theta}_{\mathrm{ML}} = \arg\max_{\boldsymbol{\theta}} \sum_{i=1}^{m} \log p_{\mathrm{model}}\left(\boldsymbol{x}^{(i)}; \boldsymbol{\theta}\right) =$$

$$= \arg\max_{\boldsymbol{\theta}} \mathbb{E}_{\mathbf{x}\sim\hat{p}_{\mathrm{data}}} \log p_{\mathrm{model}}\left(\boldsymbol{x}; \boldsymbol{\theta}\right)$$

It is empirically known from the observed data → empirical distribution

Maximum Likelihood Estimation

- The empirical distribution is derived directly from observed data points without making assumptions about the underlying probability distribution.
- Through empirical observation and data collection, we can build a distribution that reflects the frequency of occurrence of different values in our dataset.
- MLE attempts to find parameters that make the model distribution as close as possible to this empirical distribution observed in the data.

Maximum Likelihood Estimation

- We can assume that we want to minimize the difference between the empirical distribution on the data and of the model

Maximum Likelihood Estimation

- We can assume that we want to minimize the difference between the empirical distribution on the data and of the model in terms of their D_{KL}

$$D_{\mathrm{KL}}\left(\hat{p}_{\mathrm{data}} \| p_{\mathrm{model}}\right) = \mathbb{E}_{\mathbf{x} \sim \hat{p}_{\mathrm{data}}}\left[\log \hat{p}_{\mathrm{data}}(\boldsymbol{x}) - \log p_{\mathrm{model}}(\boldsymbol{x}; \boldsymbol{\theta})\right] =$$

does not depend on parameters of the model

Maximum Likelihood Estimation

- We can assume that we want to minimize the difference between the empirical distribution on the data and of the model in terms of their D_{KL}

$$D_{\mathrm{KL}}\left(\hat{p}_{\mathrm{data}} \| p_{\mathrm{model}}\right) = \mathbb{E}_{\mathbf{x} \sim \hat{p}_{\mathrm{data}}}\left[\log \hat{p}_{\mathrm{data}}(\boldsymbol{x}) - \log p_{\mathrm{model}}(\boldsymbol{x}; \boldsymbol{\theta})\right] =$$

$$= -\mathbb{E}_{\mathbf{x} \sim \hat{p}_{\mathrm{data}}}\left[\log p_{\mathrm{model}}(\boldsymbol{x}; \boldsymbol{\theta})\right] =$$

$$= -\sum_{\mathbf{x} \sim \hat{p}_{\mathrm{data}}} \hat{p}_{\mathrm{data}}(\boldsymbol{x}; \boldsymbol{\theta}) \log p_{\mathrm{model}}(\boldsymbol{x}; \boldsymbol{\theta})$$

Cross-entropy of the model distribution with respect to data one

Maximum Likelihood Estimation

- MLE can also be extended to conditional probabilities p(***y*** | ***x***;***θ***) which represent what is estimated by classifiers
- We are under the basic machine learning hypothesis on the data, namely that they are independent and identically distributed (i.i.d. assumption)

$$\boldsymbol{\theta}_{\mathrm{ML}} = \arg\max_{\boldsymbol{\theta}} P(\boldsymbol{Y}|\boldsymbol{X};\boldsymbol{\theta}) =$$

i.i.d. hypothesis

$$= \arg\max_{\boldsymbol{\theta}} \sum_{i=1}^{m} \log P(\boldsymbol{y}^{(i)}|\boldsymbol{x}^{(i)};\boldsymbol{\theta})$$

Maximum Likelihood Estimation

- It can be proven that the MSE minimization correspondes to MLE maximization if the data distribution is gaussian

- The hypothesis is that the model makes an estimation $\hat{y}(\boldsymbol{x}, \boldsymbol{w})$ from the data of the mean of the gaussian but not the variance

From frequentist statistics…

- MLE represents the approach to maximizing the estimate in the perspective of a frequentist statistics
  - The estimate of parameter $\boldsymbol{\theta}$ is a statistical variable that is a function of the data, which are also seen as random
  - The uncertainty on the estimate is represented by its variance

… to Bayesian statistics

- In Bayesian statistics, on the contrary, the data are considered as already observed and are not random while $\boldsymbol{\theta}$ is exact, but not known
- Our imperfect knowledge of $\boldsymbol{\theta}$ is represented by the prior, that is, its probability distribution $p(\boldsymbol{\theta})$ before observing the data
- The choice of the prior is crucial as it should represent our knowledge over $\boldsymbol{\theta}$ and should be informative
- In this context arise the concept of Maximum A Posteriori Estimation (MAP)

Maximum A Posteriori Estimation

- The estimation process uses Bayes' rule that estimates a *posterior probability distribution on* $\boldsymbol{\theta}$

posterior *likelihood* *prior*

$$p\left(\boldsymbol{\theta}|x^{(1)},\ldots,x^{(m)}\right) = \frac{p\left(x^{(1)},\ldots,x^{(m)}|\boldsymbol{\theta}\right)p(\boldsymbol{\theta})}{p\left(x^{(1)},\ldots,x^{(m)}\right)}$$

Constant since data are known

Maximum A Posteriori Estimation

- In this context, the choice of the prior is important:
  - It should reflect the a priori uncertainty on the value of **θ**
  - High entropy of the prior, *high information content*

posterior *likelihood* *prior*

$$p\left(\boldsymbol{\theta}|x^{(1)},\ldots,x^{(m)}\right)=\frac{p\left(x^{(1)},\ldots,x^{(m)}|\boldsymbol{\theta}\right)p(\boldsymbol{\theta})}{p\left(x^{(1)},\ldots,x^{(m)}\right)}$$

Constant since data are known

Maximum A Posteriori Estimation

- MAP estimate is the choice of a point value of $\boldsymbol{\theta}$ instead of an entire distribution

$$\boldsymbol{\theta}_{\mathrm{MAP}} = \arg\max_{\boldsymbol{\theta}} p(\boldsymbol{\theta}|\boldsymbol{x}) = \arg\max_{\boldsymbol{\theta}} \log p(\boldsymbol{x}|\boldsymbol{\theta}) + \log p(\boldsymbol{\theta})$$

  - We choose the value of $\boldsymbol{\theta}$ that maximizes the posterior
  - It is the sum of the MLE and the prior
  - The MAP estimate reduces the variance *of the estimator* compared to the MLE, but generally increases its bias
  - Useful when we have knowledge about $\boldsymbol{\theta}$ that we cannot find in the data

MLE vs MAP

MLE

- Maximize the likelihood

$$P(\mathcal{D} \mid \boldsymbol{\theta})$$

- Data distribution is not known
- May overfit over the observed data

MAP

- Maximize the posterior

$$P(\boldsymbol{\theta} \mid \mathcal{D})$$

- Incorporates prior knowledge $P(\boldsymbol{\theta})$
- Regularization thanks to the prior

MLE vs MAP – usage

MLE

- Large data set, and the effect of prior beliefs is negligible.
- Lack of reliable prior information.
- Objective is towards models grounded in frequentist statistics.

MAP

- Small or noisy data sets.
- Strong prior knowledge or domain expertise.
- Objective is towards regularization the estimate and avoid overfitting.
- Bayesian statistics.